

## Math 372: Homework 04

*Due Wednesday, Feb 11*

**Problem 1** (Exercise 2.3.34). **This problem was accidentally repeated from HW 3. You can ignore this problem.**

**Problem 2** (Exercise 2.3.39). A box in a supply room contains 15 compact fluorescent lightbulbs, of which 5 are rated 13-watt, 6 are rated 18-watt, and 4 are rated 23-watt. Suppose that three of these bulbs are randomly selected.

- (a) What is the probability that exactly two of the selected bulbs are rated 23-watt?
- (b) What is the probability that all three of the bulbs have the same rating?
- (c) What is the probability that one bulb of each type is selected?
- (d) If bulbs are selected one by one until a 23-watt bulb is obtained, what is the probability that it is necessary to examine at least 6 bulbs?

**Problem 3** (Exercise 2.4.47). Consider randomly selecting a student at a large university. Let  $A$  be the event that the selected student has a Visa card, let  $B$  be the event that the student has a MasterCard, and let  $C$  be the event that they have an American Express card. Suppose that

$$\mathbb{P}[A] = .6, \quad \mathbb{P}[B] = .4, \quad \mathbb{P}[C] = .2 \quad (1)$$

$$\mathbb{P}[A \cap B] = .3, \quad \mathbb{P}[A \cap C] = .15, \quad \mathbb{P}[B \cap C] = .1, \quad (2)$$

$$\text{and } \mathbb{P}[A \cap B \cap C] = .08. \quad (3)$$

- (a) What is the probability that the selected student has at least one of the three types of cards?
- (b) What is the probability that the selected student has both a Visa card and a MasterCard, but not an American Express card?
- (c) Calculate and interpret  $\mathbb{P}[B \mid A]$  and  $\mathbb{P}[A \mid B]$ .
- (d) If we learn that the selected student has an American Express card, what is the probability that she or he also has both a Visa card and a MasterCard?
- (e) Given that the selected student has an American Express card, what is the probability that she or he has at least one of the other two types of cards?

**Problem 4** (2.4.63). For customers purchasing a refrigerator at a certain appliance store, let  $A$  be the event that the refrigerator was manufactured in the U.S.,  $B$  be the event that the refrigerator had an icemaker, and  $C$  be the event that the customer purchased an extended warranty.

Relevant probabilities are

$$\mathbb{P}[A] = .75 \quad \mathbb{P}[B \mid A] = .9 \quad \mathbb{P}[B \mid A^c] = .8$$

$$\mathbb{P}[C \mid A \cap B] = .8 \quad \mathbb{P}[C \mid A \cap B^c] = .6$$

$$\mathbb{P}[C \mid A^c \cap B] = .7 \quad \mathbb{P}[C \mid A^c \cap B^c] = .3.$$

- (a) Construct a tree diagram consisting of first-, second-, and third-generation branches, and place an event label and appropriate probability next to each branch.
- (b) Compute  $\mathbb{P}[A \cap B \cap C]$ .
- (c) Compute  $\mathbb{P}[B \cap C]$ .
- (d) Compute  $\mathbb{P}[C]$ .

- (e) Compute  $\mathbb{P}[A \mid B \cap C]$ , the probability of a U.S. purchase given that an icemaker and extended warranty are also purchased.

**Problem 5** (Exercise 102 on p.93). A sample of 2026 students were asked about their eye color, with the following results:

|        | Blue | Brown | Green | Hazel | Total |
|--------|------|-------|-------|-------|-------|
| Male   | 370  | 352   | 198   | 187   | 1107  |
| Female | 359  | 290   | 110   | 160   | 919   |
| Total  | 729  | 642   | 308   | 347   | 2026  |

Suppose that one of these 2026 students is randomly selected. Let  $F$  denote the event that the selected individual is female, and  $A, B, C$  and  $D$  represent the events that they have blue, brown, green, and hazel eyes, respectively.

- Calculate  $\mathbb{P}[F]$  and  $\mathbb{P}[C]$
- Calculate  $\mathbb{P}[F \cap C]$ . Are these events independent? Why or why not?
- If the selected individual has green eyes, what's the probability that he or she is a female?
- If the selected individual is female, what is the probability that she has green eyes?
- What is the "conditional distribution" of eye color for females? (i.e.,  $\mathbb{P}[A|F], \mathbb{P}[B|F], \mathbb{P}[C|F], \mathbb{P}[D|F]$ ), and what is it for males? Compare the two distributions.

**Problem 6.** Two cards are randomly drawn from a deck of cards. Let

$A = [\text{at least one ace is drawn}]$

$B = [\text{both cards are aces}]$

$S = [\text{the ace of spades is drawn}]$

Compute the following conditional probabilities:

- $\mathbb{P}[S \mid B]$
- $\mathbb{P}[B \mid S]$
- $\mathbb{P}[B \mid A]$
- Note that the answer to (b) is almost twice as large as the answer to (c). Do you find this result surprising?

**Problem 7.** Recall the following definition:

Events  $A$  and  $B$  are **independent** if and only if  $\mathbb{P}[A \cap B] = \mathbb{P}[A]\mathbb{P}[B]$

Assume that  $A$  and  $B$  are independent. Explain why  $A^c$  and  $B$  must be independent, and also why  $A^c$  and  $B^c$  must be independent.

**Problem 8** (Exercise 2.5.71). A space mining company currently has two active projects, one on the moon and one in the Kuiper belt. Let  $L$  be the event that the lunar mining project is successful, and let  $K$  be the event that the Kuiper belt mining project is successful. Suppose that  $L$  and  $K$  are independent events with  $\mathbb{P}[L] = 0.4$  and  $\mathbb{P}[K] = 0.7$ .

- (a) If the lunar project is not successful, what is the probability that the Kuiper belt project is also not successful? Explain your reasoning.
- (b) What is the probability that at least one of the two mining projects will be successful?
- (c) Given that at least one of the two projects is successful, what is the probability that only the lunar project is successful?

**Problem 9** (Exercise 2.5.77). Five explosive devices are placed underneath the walls of the enemy stronghold, and are set to detonate simultaneously. Each bomb has a 4% chance of failing to detonate at the specified time. Assuming that the probabilities are independent, calculate  $\mathbb{P}[\text{at least one detonation occurs}]$  and  $\mathbb{P}[\text{at least one bomb fails to detonate}]$

**Problem 10** (Similar to Example 2.31). An email provider observes that 50% of emails are spam. It intends to filter spam emails before they reach the inbox. The filter's accuracy for detecting spam email is 99% and chances of tagging a non-spam email as spam is 5%.

- (a) Draw a tree diagram to illustrate this situation
- (b) If a certain email is tagged as spam, find the probability that it is not a spam email. That is, find

$$\mathbb{P}[\text{email is not spam} \mid \text{email is tagged as spam}]$$

**Problem 11.** Consider a universe where Netflix has exactly 6 shows, numbered 1 through 6. Suppose we randomly select a subscriber and ask which shows they watch. For each  $i = 1, 2, \dots, 6$ , let  $E_i = \{i\}$  be the event that the subscriber watches show  $i$ . Suppose that

$$P(E_1) = P(E_6) = .10, \quad P(E_2) = P(E_5) = .15, \quad P(E_3) = P(E_4) = .25.$$

Define events  $A, B, C$  by

$$A = \{2, 4, 6\}, \quad B = \{1, 2, 3\}, \quad C = \{2, 3, 4, 5\}, \quad D = \{1, 3, 5\}.$$

- (a) Are  $A$  and  $B$  dependent or independent?
- (b) Are  $A$  and  $C$  dependent or independent?
- (c) Are  $A$  and  $D$  dependent?
- (d) Can two mutually exclusive events  $A$  and  $B$  with  $P(A) > 0$  and  $P(B) > 0$  be independent? Why or why not?

**Problem 12.** The prevalence of bird flu in a wild bird population is 1%. A new diagnostic test is advertised as having a false negative rate of 0.03 and a false positive rate of 0.05. To be precise, this means that

$$\mathbb{P}[\text{test is negative} \mid \text{bird has the flu}] = 0.03$$

$$\mathbb{P}[\text{test is positive} \mid \text{bird doesn't have the flu}] = 0.05$$

- (a) What is the probability that a randomly-selected bird tests positive? (*You don't need to simplify your answer.*)
- (b) Suppose that a randomly-selected bird tests positive. What is the probability that the bird has the disease? (*You don't need to simplify your answer.*)

**Problem 13** (Rolling two dice). Two 6-sided dice are rolled and added together. Let  $X$  be the sum.

- (a) What is the probability mass function of  $X$ ? [*Hint*: First determine  $p(1), p(2), \dots$  and so on.]
- (b) Determine the cumulative density function of  $X$  and graph it.
- (c) What is the expected value of  $X$ ?

**Problem 14** (Rolling a d6 ‘with advantage’). In the game *Dungeons & Dragons*, there is a dice-rolling mechanic called “rolling with advantage” in which the player gets to roll a dice twice and keep the largest value. This bonus is typically regarded as a good thing, as higher rolls tend to be better in the game. In this problem, we’ll investigate how significant this bonus is. Two 6-sided dice are rolled. Let  $M$  be the maximum of the two rolls.

- (a) What is the probability mass function of  $M$ ? [*Hint*: First determine  $p(1), p(2), \dots$  and so on.]
- (b) Determine the cumulative density function of  $M$  and graph it.
- (c) What is the expected value of  $M$ ?
- (d) Suppose you are given the choice between two magical weapons: a magical sword that deals  $1d6 + 1$  damage, and a magical spear that deals  $1d6$  damage *with advantage*. Which magical weapons has a higher expected damage? (*Here,  $1d6$  means “roll one six-sided dice”.*)